

CSE 344 - SYSTEM PROGRAMMING

FINAL PROJECT

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I/O MULTIPLEXING DESIGN

For this issue, select function was used. The descriptor of server and the non - zero descriptors in the clients array are attached in fd_set. If there are readable or accessible descriptors, or five seconds has been passed, select function returns a value. If status is zero, that means occurring broadcast timeout. fd_set is checked, if server_fd exists in the set, new clients are accepted. If any client fd exists in the set, that means that a message comes.

For timeout mechanism, time function was used. Firstly current time is taken and remaining time is set as 5 seconds. At every iteration, remaining time is updated according to passed time, and remaining time is sent to select function as a parameter. If remaining is 0 or non-positive, then remaining is set as 5 seconds again. After 5 seconds, price broadcast is sent to network directly.

The udp descriptor was not added to fd_set, since server was going to send broadcast to ticker only, so it is not necessary.

PER-CLIENT STATE

```
40  typedef struct {
41      int fd;
42      char username[32];
43      char type[16];
44      char line_buf[512]; /* partial read buffer */
45      char *tok_buf;
46      stock_t *portfolio;
47      int size;
48      int cnt;
49      int start;
50  } client_t;
```

```
typedef struct {  
    char symbol[SYMBOL_SIZE];  
    int quantity;  
} stock_t;
```

For portfolio data structure, stock_t type was used. Memory as much as the size of the global stock is allocated. If client buys new stock, size is increased. If the stock exists already in the portfolio, size is not increased and stock is updated. The size is the number of stock types in the portfolio.

tok_buf is for line by line message handling. If the user sends messages respectively, these messages can be detected as one line because of tcp paradigm. For this situation, whole message is not sent to message_handling function, instead, the message is tokenized with '\n' delimiter, and splitted commands are sent via tok_buf pointer.

cnt is a counter used for partial_read prevention. Start is a flag used to handle the commands sent before JOIN message. If start is zero, then it means that JOIN was not called.

PRICE UPDATE MECHANISM

Firstly, price of the stock in the global market is updated. If the symbol is not in the global market, error message is given. If not here any error, the portfolio of the client is updated.

Since it is wanted that prices is reported and stored with 2 decimal places, prices were stored as integer numbers. When it is printed, it is divided by 100. Therefore prices had been stored with two decimal places. Normally, the formula was

$price = price + qty * 0.01$

$price = price - qty * 0.01$

The new formula is

$price = price + qty$

$price = price - qty$

After the reporting, without any waiting, the price broadcast is sent to ticker.

PARTIAL READ/WRITE HANDLING

Partial Read Handling:

Firstly, standard read function is used. The read buf is appended to line_buf and it is checked if it includes new line character, or not. If not, the next client is assessed. If it includes '\n', whole line_buf is sent to message_handling function line by line tokenizing.

Partial Write Handling:

For this matter, special_write function was implemented. Until the wanted amount of size bytes are written, the simple write operation goes on.

```
void special_write(int fd, char *buf, int size) {  
    int written_byte;  
  
    while(1) {  
        written_byte = write(fd, buf, size);  
  
        if(written_byte <= 0) {  
            if(errno == EINTR) continue;  
            break;  
        }  
  
        if(written_byte == size) break;  
  
        buf += written_byte;  
        size -= written_byte;  
    }  
}
```

SIGNAL HANDLING AND SHUTDOWN

Server → Ctrl – C:

- 1) The signal arrives.
- 2) The flow of control passes to signal handling function.
- 3) Flag “done” is set to 1.
- 4) Select wakes up because of the signal. Returned value is -1 and errno is EINTR.
- 5) Flag done is checked. If done is 1, shutdown message is sent, and general descriptors are closed and the loop is broken.
- 6) In the main function, clean_portfolios function is called, and portfolios are freed.
- 7) Other blocks are freed.

Client → Ctrl – D

- 1) EOF is typed from the STDIN.
- 2) fgets returns NULL pointer, and feof function returns true.
- 3) QUIT message is sent to the server.
- 4) Flag done is set to 1.
- 5) The descriptor is closed.
- 6) The loop is broken.

TEST SCENARIOS

Scenario 1:

Server:

```
muhammed@debian:~/Desktop/230104004017$ ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=7 ip=127.0.0.1
[SERVER] JOIN username=bob type=TRADER fd=7
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=AAPL qty=100 old_price=150.00 new_price=151.00
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=bob symbol=AAPL qty=50 old_price=151.00 new_price=151.50
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=alice reason=QUIT
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=bob reason=QUIT
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
muhammed@debian:~/Desktop/230104004017$
```

Trader: alice

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY AAPL 100
[TRADER alice] SENT BUY AAPL 100
[TRADER alice] RECEIVED OK BUY AAPL 100 151.00
PORTFOLIO
[TRADER alice] SENT PORTFOLIO
[TRADER alice] RECEIVED OK PORTFOLIO AAPL:100
[TRADER alice] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$
```

Trader: bob

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 bob
[TRADER bob] CONNECTED server=127.0.0.1:8080.
[TRADER bob] SENT JOIN
[TRADER bob] RECEIVED OK JOIN bob
BUY AAPL 50
[TRADER bob] SENT BUY AAPL 50
[TRADER bob] RECEIVED OK BUY AAPL 50 151.50
PORTFOLIO
[TRADER bob] SENT PORTFOLIO
[TRADER bob] RECEIVED OK PORTFOLIO AAPL:50
[TRADER bob] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$ █
```

Scenario 2:

Server:

```
muhammed@debian:~/Desktop/230104004017$ ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=7 ip=127.0.0.1
[SERVER] JOIN username=bob type=TRADER fd=7
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=8 ip=127.0.0.1
[SERVER] JOIN username=Mehmet type=TRADER fd=8
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=9 ip=127.0.0.1
[SERVER] JOIN username=Mert type=ANALYST fd=9
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=10 ip=127.0.0.1
[SERVER] JOIN username=Ali type=ANALYST fd=10
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=GOOGL qty=45 old_price=2800.00 new_price=2800.45
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] SELL trader=alice symbol=GOOGL qty=33 old_price=2800.45 new_price=2800.12
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=bob symbol=AMZN qty=222 old_price=3400.00 new_price=3402.22
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] SELL trader=bob symbol=AMZN qty=45 old_price=3402.22 new_price=3401.77
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=Mehmet symbol=AAPL qty=12 old_price=150.00 new_price=150.12
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] SELL trader=Mehmet symbol=AAPL qty=3 old_price=150.12 new_price=150.09
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=Ali reason=QUIT
[SERVER] CLIENT_DISCONNECTED username=Mert reason=QUIT
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=Mehmet reason=QUIT
[SERVER] CLIENT_DISCONNECTED username=bob reason=QUIT
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=alice reason=QUIT
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
muhammed@debian:~/Desktop/230104004017$
```

Trader 1:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY GOOGL 45
[TRADER alice] SENT BUY GOOGL 45
[TRADER alice] RECEIVED OK BUY GOOGL 45 2800.45
SEND GOOGL 23
[TRADER alice] SENT SEND GOOGL 23
[TRADER alice] RECEIVED ERR UNKNOWN SEND GOOGL 23
SELL GOOGL 33
[TRADER alice] SENT SELL GOOGL 33
[TRADER alice] RECEIVED OK SELL GOOGL 33 2800.12
[TRADER alice] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$ █
```

Trader 2:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 bob
[TRADER bob] CONNECTED server=127.0.0.1:8080.
[TRADER bob] SENT JOIN
[TRADER bob] RECEIVED OK JOIN bob
BUY AMZN 222
[TRADER bob] SENT BUY AMZN 222
[TRADER bob] RECEIVED OK BUY AMZN 222 3402.22
SELL AMZN 45
[TRADER bob] SENT SELL AMZN 45
[TRADER bob] RECEIVED OK SELL AMZN 45 3401.77
[TRADER bob] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$ █
```

Trader 3:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 Mehmet
[TRADER Mehmet] CONNECTED server=127.0.0.1:8080.
[TRADER Mehmet] SENT JOIN
[TRADER Mehmet] RECEIVED OK JOIN Mehmet
BUY AAPL 12
[TRADER Mehmet] SENT BUY AAPL 12
[TRADER Mehmet] RECEIVED OK BUY AAPL 12 150.12
SELL 4
[TRADER Mehmet] SENT SELL 4
[TRADER Mehmet] RECEIVED ERR UNKNOWN SELL 4
SELL AAPL 3
[TRADER Mehmet] SENT SELL AAPL 3
[TRADER Mehmet] RECEIVED OK SELL AAPL 3 150.09
[TRADER Mehmet] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$ █
```


Analyst 1:

```
muhammed@debian:~/Desktop/230104004017$ ./analyst 127.0.0.1 8080 Mert
```

```
[ANALYST Mert] CONNECTED server=127.0.0.1:8080.
```

[ANALYST Mert] SENT JOIN

[ANALYST Mert] RECEIVED OK JOIN Mert

LIST

[ANALYST Mert] SENT LIST

```
[ANALYST Mert] RECEIVED OK LIST alice bob Mehmet Mert Ali
```

[ANALYST Mert] DISCONNECTED reason=QUIT

```
muhammed@debian:~/Desktop/230104004017$
```

Analyst 2:

```
muhammed@debian:~/Desktop/230104004017$ ./analyst 127.0.0.1 8080 Ali
```

```
[ANALYST Ali] CONNECTED server=127.0.0.1:8080.
```

[ANALYST Ali] SENT JOIN

[ANALYST Ali] RECEIVED OK JOIN Ali

REPORT

[ANALYST Ali] SENT REPORT
[ANALYST Ali] RECEIVED OK REPORT AAPL:150.00 GOOGL:2800.12 MSFT:380.00 TSLA:250.00 AMZN:3401.77 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AVGO:1200.00 ADBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZM:70.09 SQ:80.90 PYPL:65.40

[ANALYST Ali] DISCONNECTED reason=QUIT

```
muhammed@debian:~/Desktop/230104004017$
```

Ticker 1:

[illegible]

Ticker 2:

[illegible]

Scenario 3:

Server:

```

muhammed@debian:~/Desktop/230104004017$ ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] CLIENT_CONNECTED. fd=7 ip=127.0.0.1
[SERVER] JOIN username=bob type=ANALYST fd=7
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=AAPL qty=100 old_price=150.00 new_price=151.00
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=bob reason=QUIT
[SERVER] CLIENT_DISCONNECTED username=alice reason=QUIT
[SERVER] PRICE_BROADCAST trigger=PERIODIC
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
muhammed@debian:~/Desktop/230104004017$

```

Trader:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY AAPL 100
[TRADER alice] SENT BUY AAPL 100
[TRADER alice] RECEIVED OK BUY AAPL 100 151.00
SELL AAPL 999
[TRADER alice] SENT SELL AAPL 999
[TRADER alice] RECEIVED ERR INSUFFICIENT_SHARES
[TRADER alice] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$
```

Analyst:

```
muhammed@debian:~/Desktop/230104004017$ ./analyst 127.0.0.1 8080 bob
[ANALYST bob] CONNECTED server=127.0.0.1:8080.
[ANALYST bob] SENT JOIN
[ANALYST bob] RECEIVED OK JOIN bob
PRICE AAPL
[ANALYST bob] SENT PRICE AAPL
[ANALYST bob] RECEIVED OK PRICE AAPL 151.00
[ANALYST bob] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$ █
```

Scenario 4:

Server:

```
muhammed@debian:~/Desktop/230104004017$ ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=7 ip=127.0.0.1
[SERVER] JOIN username=bob type=ANALYST fd=7
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=G00GL qty=23 old_price=2800.00 new_price=2800.23
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=AMZN qty=34 old_price=3400.00 new_price=3400.34
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=AAPL qty=45 old_price=150.00 new_price=150.45
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_DISCONNECTED username=bob reason=QUIT
[SERVER] CLIENT_DISCONNECTED username=alice reason=QUIT
[SERVER] PRICE_BROADCAST trigger=PERIODIC
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
muhammed@debian:~/Desktop/230104004017$ █
```

```

muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY GOOGL 23
[TRADER alice] SENT BUY GOOGL 23
[TRADER alice] RECEIVED OK BUY GOOGL 23 2800.23
BUY AMZN 33
[TRADER alice] SENT BUY AMZN 33
[TRADER alice] RECEIVED ERR UNKNOWN_SYMBOL
BUY AMZN 34
[TRADER alice] SENT BUY AMZN 34
[TRADER alice] RECEIVED OK BUY AMZN 34 3400.34
BUY AAPL 45
[TRADER alice] SENT BUY AAPL 45
[TRADER alice] RECEIVED OK BUY AAPL 45 150.45
[TRADER alice] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$

```

```
top/230104004017$ ./analyst 127.0.0.1 8080 bob
D server=127.0.0.1:8080.
N
OK JOIN bob

ORT
OK REPORT AAPL:150.45 GOOGL:2800.23 MSFT:380.00 TSLA:250.00 AMZN:3400.34 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40
60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABNB:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 Roku:65.30 ZM:70.09 SQ:80.90 PYPL:65.40

T
OK LIST alice bob
CTED reason=QUIT
top/230104004017$ █
```

[illegible]

Server:

Trader 1:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
Killed
muhammed@debian:~/Desktop/230104004017$
```

Trader 2:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 bob
[TRADER bob] CONNECTED server=127.0.0.1:8080.
[TRADER bob] SENT JOIN
[TRADER bob] RECEIVED OK JOIN bob
[TRADER bob] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$
```

Trader 3:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 Mehmet
[TRADER Mehmet] CONNECTED server=127.0.0.1:8080.
[TRADER Mehmet] SENT JOIN
[TRADER Mehmet] RECEIVED OK JOIN Mehmet
[TRADER Mehmet] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$
```

Kill Operation:

```
muhammed@debian:~/Desktop/230104004017$ ps aux | grep trader
muhammed  6883  0.0  0.0  2364  556 pts/1    S+   16:26   0:00 ./trader 127.0.0.1 8080 alice
muhammed  6884  0.0  0.0  2364  556 pts/2    S+   16:26   0:00 ./trader 127.0.0.1 8080 bob
muhammed  6889  0.0  0.0  2364  556 pts/3    S+   16:26   0:00 ./trader 127.0.0.1 8080 Mehmet
muhammed  6902  0.0  0.0  6240  644 pts/4    S+   16:26   0:00 grep trader
muhammed@debian:~/Desktop/230104004017$ kill -9 2364
bash: kill: (2364) - No such process
muhammed@debian:~/Desktop/230104004017$ kill -9 6883
muhammed@debian:~/Desktop/230104004017$
```

New Trader:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
[TRADER alice] DISCONNECTED reason=QUIT
muhammed@debian:~/Desktop/230104004017$
```


Scenario 6:

Server:

```
muhammed@debian:~/Desktop/230104004017$ valgrind --leak-check=full --show-leak-kinds=all --track-fds=yes ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
==7105== Memcheck, a memory error detector
==7105== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==7105== Using Valgrind-3.16.1 and LibVEX; rerun with -h for copyright info
==7105== Command: ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
==7105==
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
==7105== Conditional jump or move depends on uninitialised value(s)
==7105==    at 0x4838969: strcat (vg_replace_strmem.c:308)
==7105==    by 0x104612: server_communication (in /home/muhammed/Desktop/230104004017/ServerTrd)
==7105==    by 0x1095E1: main (in /home/muhammed/Desktop/230104004017/ServerTrd)
==7105==
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=7 ip=127.0.0.1
[SERVER] JOIN username=bob type=TRADER fd=7
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=GOOGL qty=23 old_price=2800.00 new_price=2800.23
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
==7105==
==7105== FILE DESCRIPTORS: 3 open at exit.
==7105== Open file descriptor 2: /dev/pts/0
==7105==    <inherited from parent>
==7105== Open file descriptor 1: /dev/pts/0
==7105==    <inherited from parent>
==7105== Open file descriptor 0: /dev/pts/0
==7105==    <inherited from parent>
==7105==
==7105== HEAP SUMMARY:
==7105==    in use at exit: 0 bytes in 0 blocks
==7105==    total heap usage: 9 allocs, 9 frees, 49,640 bytes allocated
==7105==
==7105== All heap blocks were freed -- no leaks are possible
==7105==
==7105== Use --track-origins=yes to see where uninitialised values come from
==7105== For lists of detected and suppressed errors, rerun with: -s
==7105== ERROR SUMMARY: 2 errors from 1 contexts (suppressed: 0 from 0)
muhammed@debian:~/Desktop/230104004017$ valgrind --leak-check=full --show-leak-kinds=all --track-fds=yes ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
==7173== Memcheck, a memory error detector
==7173== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==7173== Using Valgrind-3.16.1 and LibVEX; rerun with -h for copyright info
==7173== Command: ./ServerTrd -p 8080 -u 8081 -s stocks.txt -l server.log
==7173==
[SERVER] SERVER_START tcp_port=8080 udp_port=8081 stocks=30
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] CLIENT_CONNECTED. fd=6 ip=127.0.0.1
==7173== Conditional jump or move depends on uninitialised value(s)
==7173==    at 0x4838969: strcat (vg_replace_strmem.c:308)
==7173==    by 0x104612: server_communication (in /home/muhammed/Desktop/230104004017/ServerTrd)
==7173==    by 0x1095E1: main (in /home/muhammed/Desktop/230104004017/ServerTrd)
==7173==
[SERVER] JOIN username=alice type=TRADER fd=6
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] BUY trader=alice symbol=AMZN qty=56 old_price=3400.00 new_price=3400.56
[SERVER] PRICE_BROADCAST trigger=TRADE
[SERVER] PRICE_BROADCAST trigger=PERIODIC
[SERVER] PRICE_BROADCAST trigger=PERIODIC
^C[SERVER] SHUTDOWN signal=SIGINT
[SERVER] CLEANUP_DONE clients=0
==7173==
==7173== FILE DESCRIPTORS: 3 open at exit.
==7173== Open file descriptor 2: /dev/pts/0
==7173==    <inherited from parent>
==7173== Open file descriptor 1: /dev/pts/0
==7173==    <inherited from parent>
==7173== Open file descriptor 0: /dev/pts/0
==7173==    <inherited from parent>
==7173==
==7173== HEAP SUMMARY:
==7173==    in use at exit: 0 bytes in 0 blocks
==7173==    total heap usage: 8 allocs, 8 frees, 49,280 bytes allocated
==7173==
==7173== All heap blocks were freed -- no leaks are possible
==7173==
==7173== Use --track-origins=yes to see where uninitialised values come from
==7173== For lists of detected and suppressed errors, rerun with: -s
==7173== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
muhammed@debian:~/Desktop/230104004017$ ps
  PID TTY          TIME CMD
  5551 pts/0    00:00:00 bash
  7226 pts/0    00:00:00 ps
muhammed@debian:~/Desktop/230104004017$
```

Trader 1 and new trader:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY GOOGL 23
[TRADER alice] SENT BUY GOOGL 23
[TRADER alice] RECEIVED OK BUY GOOGL 23 2800.23
[TRADER alice] RECEIVED SERVER SHUTDOWN
[TRADER alice] DISCONNECTED reason=shutdown
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 alice
[TRADER alice] CONNECTED server=127.0.0.1:8080.
[TRADER alice] SENT JOIN
[TRADER alice] RECEIVED OK JOIN alice
BUY AMZN 56
[TRADER alice] SENT BUY AMZN 56
[TRADER alice] RECEIVED OK BUY AMZN 56 3400.56
[TRADER alice] RECEIVED SERVER SHUTDOWN
[TRADER alice] DISCONNECTED reason=shutdown
muhammed@debian:~/Desktop/230104004017$ ps
  PID TTY          TIME CMD
  5555 pts/1    00:00:00 bash
  7227 pts/1    00:00:00 ps
muhammed@debian:~/Desktop/230104004017$ █
```

Trader 2:

```
muhammed@debian:~/Desktop/230104004017$ ./trader 127.0.0.1 8080 bob
[TRADER bob] CONNECTED server=127.0.0.1:8080.
[TRADER bob] SENT JOIN
[TRADER bob] RECEIVED OK JOIN bob
[TRADER bob] RECEIVED SERVER SHUTDOWN
[TRADER bob] DISCONNECTED reason=shutdown
muhammed@debian:~/Desktop/230104004017$ ps
  PID TTY          TIME CMD
  5557 pts/2    00:00:00 bash
  7228 pts/2    00:00:00 ps
muhammed@debian:~/Desktop/230104004017$ █
```

Ticker:

```
muhammed@debian:~/Desktop/230104004017$ ./ticker 8081
[TICKER] LISTENING udp port=8081
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
[TICKER] RECEIVED PRICE UPDATE AAPL:150.00 GOOGL:2800.00 MSFT:380.00 TSLA:250.00 AMZN:3400.00 META:300.50 NVDA:850.25 NFLX:600.00 INTC:45.10 AMD:110.20 IBM:135.50 ORCL:115.75 CSCO:50.25 QCOM:140.80 TXN:160.40 AV
GO:1200.00 ABBE:550.60 CRM:280.89 SAP:180.30 NOW:750.00 INTU:620.40 UBER:75.25 ABBN:155.50 SPOT:290.80 SNOW:160.19 PLTR:24.50 ROKU:65.30 ZH:70.09 5Q:80.90 PYPL:65.40
muhammed@debian:~/Desktop/230104004017$ ps
  PID TTY          TIME CMD
  5559 pts/3    00:00:00 bash
  7229 pts/3    00:00:00 ps
muhammed@debian:~/Desktop/230104004017$ █
```


CHALLENGES

Partial Reading / Writing Problem:

For this problem, firstly, `special_write` and `special_reading` function were implemented. There was not any problem for `special_write` function but there was a problem in `special_read` function. Actually, this error was detected by AI.

For every message, new line character should exist, but the sent by the server messages were not including '\n'. Because clients wait the '\n', the program locked in `special_read` function.

At the begining, for every replies from server, '\n' was attached. Therefore, lock problem was solved. However, there was another problem. `special_read` function was violating the select function logic, since if new line character is not typed for a long time, the program itarates through the while loop in `special_read` function, and other available file descriptors are held. Essentially, `line_buf` was not being used correctly, and logically. Instead of it, read operation was handled in the select loop. If therrre is no any '\n', the loop continues. When a new line character exists, the `line_buf` is tokenized and sent to `message_handling` function.

UDP Broadcast Problem:

The cause of this problem was actually that I did not understand completely bind function logic.

When any message arrives in the machine, operating system looks the port of the message. What programs listen that port? Those programs should have state this notification to os. Since ticker clients takes the messages from the server, it should inform the os. My problem was about this issue. It was solved by using bind function in correct place.

Actually, these problems and some others was detected by AI. In this process, AI provides some solutions and informations.